

JEE MAIN REVISION MASTER: ELECTROCHEMISTRY 2026 4 QUESTIONS 6.67% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 35 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Electrochemistry - Core Application	35 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Electrochemistry - Core Application (Very High Yield — 35 PYQs)

Core Principle: Key problem pattern in Kcet Electrochemistry.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The conductivity of centimolar solution of KCl at 298 K is $0.021 \text{ Ohm}^{-1} \text{ cm}^{-1}$ and the resistance of the cell containing ..." [Ans: C]
- [2026 Shift] Q#2 (MCQ): "Given below are the half-cell reactions: $\text{Mn}^{2+} + 2\text{e}^{-} \rightarrow \text{Mn}$ ($E^{\circ} = -1.18 \text{ V}$) $\text{Mn}^{3+} + \text{e}^{-} \rightarrow \text{Mn}^{2+}$ ($E^{\circ} = 1.50 \text{ V}$) ... " [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): " $\Delta_{\text{m}}(\text{NH}_4\text{OH})^{\circ}$ is equal to _____" [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The conductivity of centimolar solution of KCl at 298 K is $0.021 \text{ Ohm}^{-1} \text{ cm}^{-1}$ and the resistance of the cell containing the solution at 298 K is 60Ω . The value of cell constant (G^*) is _____.

- (A) 3.28 cm^{-1}
- (B) 1.26 cm^{-1}
- (C) 3.34 cm^{-1} **[Correct]**
- (D) 1.34 cm^{-1}

Correct Answer: C

Solution Analysis: Conductivity is related to conductance and cell constant by $\kappa = G \times G^*$ where $G = (1 / R)$ So, $\kappa = (1 / R) \times G^*$ This gives $G^* = \kappa \times R$ Given, $\kappa = 0.021 \text{ Ohm}^{-1} \text{ cm}^{-1}$ and $R = 60 \Omega$ So, $G^* = 0.021 \times 60$ $G^* = 1.26 \text{ cm}^{-1}$ So, the correct option is 1.26 cm^{-1} Option B.